Photonic Crystal Fiber Loop Mirror-Based Chemical Vapor Sensor

Luo Niu, Chun-liu Zhao, Liang Qi, Chi Chiu Chan, Juan Kang, Shangzhong Jin, Jiangtao Guo, and Huifeng Wei

***Abstract*—A compact chemical vapor sensor based on a fiber loop mirror (FLM) inserted with a high-birefringent photonic crystal fiber (HiBi-PCF) is proposed and experimentally demonstrated. Because the resonant dips of the HiBi-PCF FLM are sensitive to the phase difference between two counter-propagating waves in the FLM, a highly sensitive chemical vapor can be achieved by measuring the resonant dip shifts when chemical vapor diffuses into the air holes of the HiBi-PCF. Experimental results show a high sensitivity of 15.5 pm/ppm for ethanol is obtained when a 5.1 cm long HiBi-PCF is used in the FLM. Three different mechanisms for the high sensitivity of the proposed sensor are discussed. A liquefaction model with a full-vector finite element method is proposed and is regarded as the main contribution.**

***Index Terms*—Optical fiber applications, optical sensors, volatile organic compounds.**

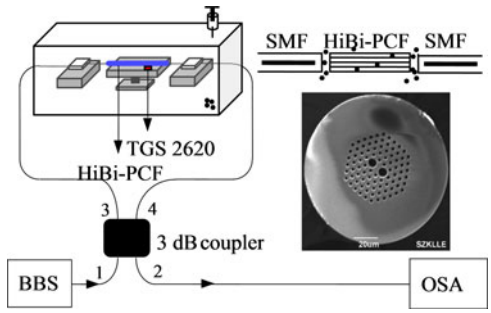

Fig. 1. Experimental setup for chemical vapor sensor based on a FLM. Insertion: the details of the sensor head and the SEM of the used HiBi-PCF. The dots represent VOC molecules.

I. Introduction

THE monitoring of volatile organic compounds (VOCs) is becoming increasingly important because most of VOCs may cause significant threaten to human health [1]. Among all the detecting methods, sensors based on optical fibers have attracted much attention in recent years due to their unique advantages such as small size, low cost and remote measurement.

Various optical fiber sensors have been demonstrated for monitoring gases and VOCs. The most common is based on optical spectrum analysis, in which the target gases or VOCs are detected by testing their optical absorption properties [2], [3]. Besides that, optical fiber sensors based on a special sensitive layer with a suitable sensing mechanism have also been proposed for VOCs and gases detecting in the past few years, such as fiber gratings [4]–[6], tapered fibers [7], the simplified photonic crystal fibers [8] and so on [9], [10]. However, these sensors always need some processes to the fiber, such as tapering the fibers, etching or side polishing the cladding of the fibers [11]. In recent years, a new kind of VOCs sensors based on an interferometer with a relatively high sensitivity have been proposed [12], [13].

Fiber loop mirror (FLM), consisting of a highly birefringent photonic crystal fiber (HiBi-PCF) combined with a 3 dB coupler, is cheap and stable, and has been widely used in sensing [14], [15], [18] and communications areas [16]. In this paper, we demonstrate a novel chemical vapor sensor by inserting a short HiBi-PCF into a FLM without splicing. As is known, the resonant dips of the HiBi-PCF FLM are sensitive to the phase difference between two counter-propagating waves in the FLM. So an extremely high sensitivity to chemical vapor can be realized by measuring the wavelength shift of the resonant dips when chemical vapor diffusing into the air holes of the HiBi-PCF. The sensor can work with a common broadband light source, without any pre-process to the fiber. Experimental results show that the sensitivity reaches up to 15.5 pm/ppm for ethanol when a 5.1 cm long PCF is used in the FLM. Three different possible mechanisms of the proposed sensor are also discussed in this paper, in which a liquefaction model with a full-vector finite element method (FEM) is proposed. The simulation results indicate that the ethanol liquefaction contributes largely to the high sensitive of the proposed sensor.

II. Experimental Setup and Principle

The proposed chemical vapor sensor, as shown in Fig. 1, consists of a 3 dB coupler and a short HiBi-PCF. The HiBi-PCF is fixed on a stable stage with a V-groove, and its both ends are aligned with two single mode fibers (SMFs), leaving a very short air gap (AG) to allow the molecules of VOC entering. The SMFs are fixed on two 3-D translation stages respectively, and both ends of the HiBi-PCF and SMFs are cleaved with a standard

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cleaving machine. All the stages are plated in a sealed acrylic glass chamber, and the concentration of VOCs in the chamber is controlled by pouring ethanol into the chamber with a pipette, whose resolution is 0.05 ml. A TGS 2620 (FIGARO Company) sensor head, based on a metal oxide semiconductor, is sensitive to VOC and can be applied as an organic vapor detector. We also placed this TGS 2620 sensor head in the chamber for the calibration of the concentration in our experiments. A broadband SLED source with 200 nm wavelength range is used as an input light source. The transmission spectra of the FLM are measured by an optical spectrum analyzer (OSA).

The main FLM principle is summarized as following. Input light is splitted by the 3 dB coupler equally into two counter-propagating waves, which subsequently recombine at the coupler after propagating around the loop. These two waves produce an optical path difference after transmitting through the HiBi-PCF because of its birefringence. Therefore, an interference response with many dips generates, which is approximately a periodic function of the wavelength, given as [15]

$$T = (1 - \cos\theta)/2 \tag{1}$$

where $\theta = 2\pi BL/\lambda$ is the phase difference of the FLM. λ is the wavelength, $B = n_x - n_y$ is the birefringence of the HiBi-PCF, n_x and n_y are the effective refractive index (RI) at the slow and fast axes, L is the length of the HiBi-PCF. The resonant dip wavelengths satisfies $2\pi BL/\lambda_{\rm dip} = 2k\pi$, where k is any integer. Thus, the resonant dip wavelengths can be described as

$$\lambda_{\rm dip} = BL/k \tag{2}$$

Here, the HiBi-PCF, which is inserted into a FLM, works as a chemical vapor sensing head. When the chemical vapors diffuse into the air holes of the HiBi-PCF through the AG, the birefringence of the HiBi-PCF B changes, while the length of the HiBi-PCF L remains the same due to its insensitivity to temperature. Therefore, the relationship between the dip wavelength change $\Delta\lambda_{\rm dip}$ and the birefringence change ΔB can be simply expressed as

$$\Delta\lambda_{\rm dip} = \Delta BL/k. \tag{3}$$

The HiBi-PCF used in our experiment is provided by Yangtze Optical Fiber and Cable Company, and the cross-sectional scanning electron micrograph (SEM) is shown in the insertion of Fig. 1. The diameters of the bigger and smaller holes are 7 and 3.2 μm, respectively, and the pitch length between centers of two adjacent holes is 5.46 μm. Here, a $\sim$5.1 cm length of HiBi-PCF is alighted with two SMFs without splicing as a chemical vapor sensing head.

III. Experiment and Results

In our experiments, ethanol is used as the chemical vapor sample because it's easy to evaporate and the vapor is harmless for experimenters. The ethanol is injected into the chamber by 0.05 ml every time, and the corresponding ethanol vapor concentration (EVC) in the chamber is changed by 250 ppm each time. When the response of the sensor reaches stable, the output spectrum will be recorded and more ethanol will be injected. Fig. 2 shows the transmission spectrum of the HiBi-PCF FLM when the EVC is 0 ppm. Several resonant dips of the FLM can be observed in the wavelength ranging from 1450 to 1600 nm. In order to observe the wavelength shift, two main dips are chosen during our experiments, one of which is at the wavelength about 1548 nm (dip A) while the other is at about 1588 nm (dip B). The wavelength spacing between the two dips is $\sim$40 nm. Meanwhile, there are also many interference dips with high frequency on the transmission spectrum due to the light interference in the two AGs.

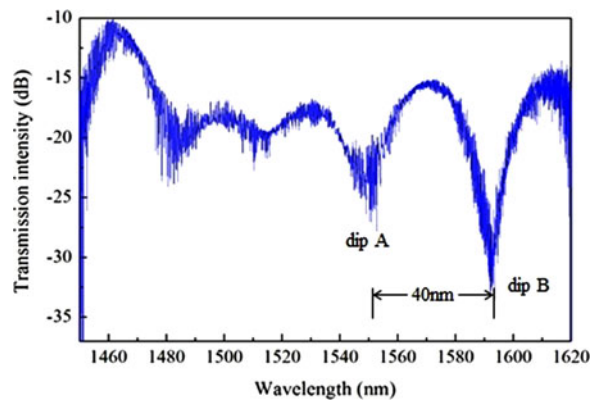

Fig. 2. Transmission spectrum of the HiBi-PCF FLM when the EVC in the chamber is 0 ppm.

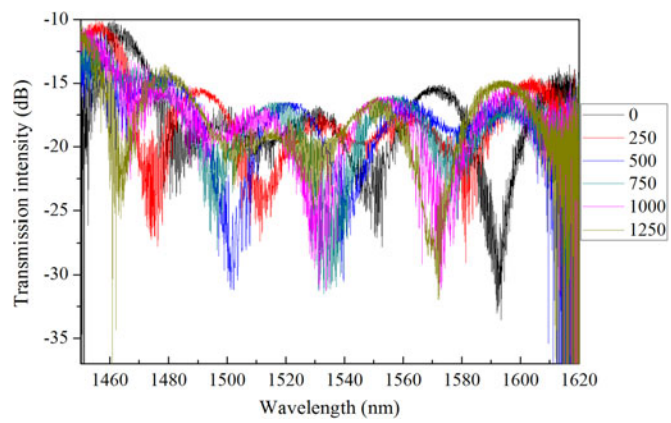

Fig. 3. Transmission spectrum of the HiBi-PCF FLM without filtering at each stable situation from 0 to 1250 ppm.

Fig. 3 shows the transmission spectra of the HiBi-PCF FLM when the response of the sensor is stable at 0, 250, 500, 750, 1000, 1250 ppm, respectively. It is obviously that both dip A and dip B have a blue-shift when the EVC increases. This is because the birefringence is inversely proportional with the increasing EVC in the air holes of the HiBi-PCF. Fig. 4 shows the transmission spectra at each stable situation after the signal denoising, filtering the resonant dips with the high frequency. Apart from the evidently blue-shift of dip A and dip B, the intensities of both dips also change as the EVC increases. The experimental relationships between EVC and the resonant wavelength of dip A and dip B are also got, as shown in Fig. 5. The fitting curves

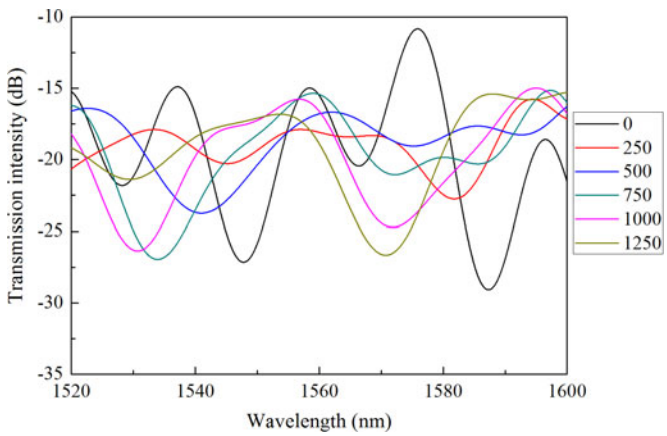

Fig. 4. Transmission spectra of the HiBi-PCF FLM at each stable situation after filtering.

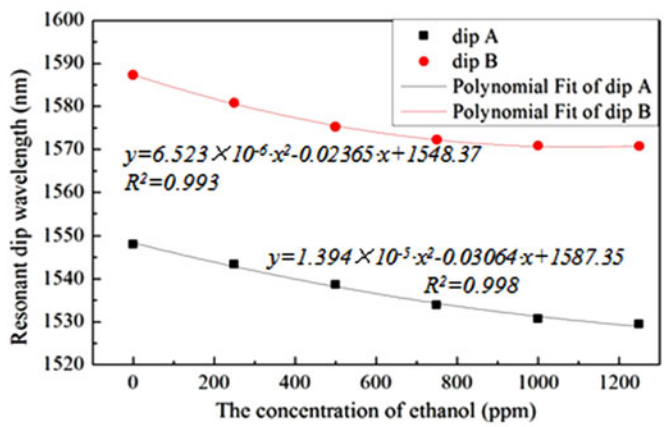

Fig. 5. The relationships between EVC and the resonant wavelength of dip A and dip B.

can be expressed as

$$y = 6.523 \times 10^{-6} \cdot x^2 - 0.02365 \cdot x + 1548.37$$

for dip A while

$$y = 1.394 \times 10^{-5} \cdot x^2 - 0.03064 \cdot x + 1587.35$$

for dip B. The experimental EVC sensitivity of dip A and dip B are ~15.5 and ~13.2 pm/ppm, respectively. Compared with other previously reported sensors, the sensitivity of the proposed sensor is considerably higher. For instance, in Ref. [13], the volume of voids responsible for the shift is ~520 pL and the measurement range is $\sim 3.7 \times 10^8$ ppm for THF. The FBG interrogator used in Ref. [13] has a resolution of 10 pm, whereas the sensitivity can be calculated to $\sim 2.7 \times 10^{-8}$ pm/ppm, which is in low sensitivity. When an OSA with 20 pm resolution is used, the EVC's resolutions are separately ~1.3 and ~1.5 ppm in the range of 0 to 1250 ppm.

The response characteristic of the proposed sensor is also been investigated. Fig. 6 shows the measured dip A as a function of time when the EVC in the chamber changes from 0 to 1250 ppm, increased by 250 ppm each time. As shown in Fig. 6, the dip

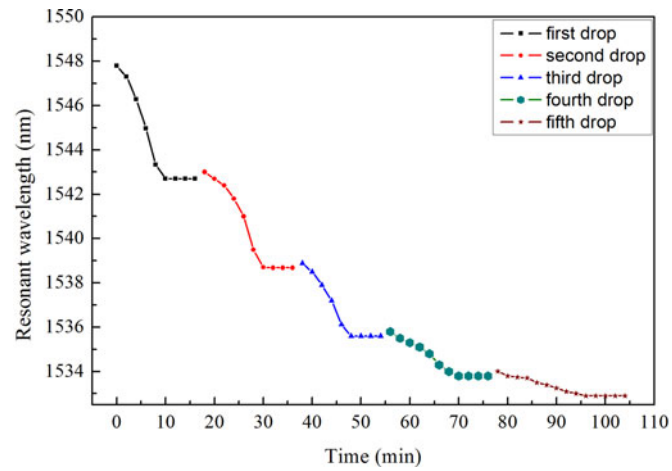

Fig. 6. The relationship between the measured dipA and time when the EVC in the chamber changes from 0 to 1250 ppm.

shift becomes weaker while the stability time becomes longer as the EVC increases. What is interesting is that the dip wavelength has a little red-shift at the first 1 ~ 2 min after injecting ethanol. This may be caused by the evaporation process of the ethanol liquid, which will be discussed in Section IV.

In order to shorten the response time of the proposed sensor, a shorter HiBi-PCF with larger air holes is better, which needs less time to reach stable when the VOC cells diffuse. Moreover, a bigger birefringence change ΔB can be considered according to Eq. (3) for a higher sensitivity of the proposed sensor.

IV. Discussion

A chemical vapor sensor based on a photonic crystal fiber interferometer with a high sensitivity has been previously reported [13]. It is known that the wavelength shift of the PCF interferometer is in proportion to birefringence changes ΔB directly. However, there is no detail analysis for why the sensitivity is so high. Here, we should discuss three major facts that may influence ΔB of the HiBi-PCF in our proposed vapor sensor, and mention that both the influence of gaseous ethanol vapor RI variation and AGs are not the main facts contributing to the high sensitivity. Instead, the ethanol liquefaction model we propose contributes largely to ΔB, which may be the main reason causing the wavelength shift largely.

A. The Influence of Gaseous Ethanol Vapor RI Variation

The chemical vapor has a very low RI, so the RI change induced by VOC is a tiny variation. Although the liquid ethanol has a RI of 1.36, the RI of gaseous ethanol vapor is as low as 1.000878. The relationship between the RI in the fiber air holes and the birefringence can be calculated by using a full-vector FEM. The birefringence of the HiBi-PCF, simulated at 1550 nm, decreases linearly with the RI in the air holes rising, as shown in Fig. 7. The trend can be expressed as $y = -0.0013 \cdot x + 0.0022$. The mode fields of the two orthogonal polarizations are shown in the insertion of Fig. 7. The birefringence of the HiBi-PCF is 9×10^{-4} when the RI is 1, which means the EVC

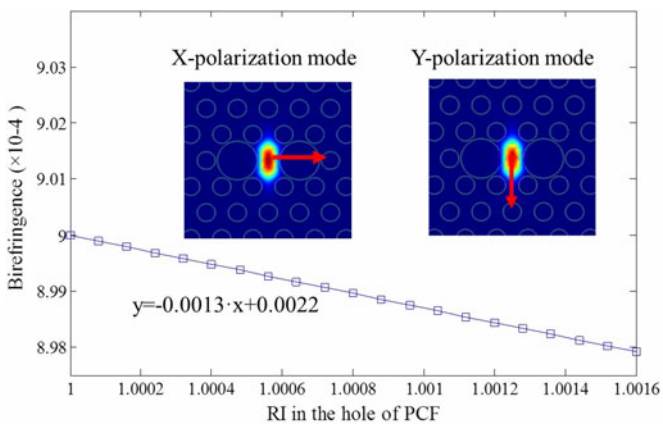

Fig. 7. The RI in the air holes of the HiBi-PCF and the birefringence of the HiBi-PCF in theory. Insertion: X and Y polarization mode fields of the HiBi-PCF at 1550 nm and room temperature.

is 0 ppm. And the birefringence will decrease to 8.9896×10^{-4} when the RI changes to 1.0008, assuming the air holes are full of gaseous ethanol vapor cells. The total birefringence change is only 1.04×10^{-6} ranging from 1 to 1.0008. According to Eq. (3), the dip wavelength has a very little shift.

Therefore, the EVC-induced birefringence change may not be considered as the main reason that contributes to the high sensitivity.

B. The Influence of two AGs

As mentioned above, the HiBi-PCF is fixed on a stable stage with a V-groove, whose both ends are aligned with SMFs without splicing. Thus, two AGs are introduced in the proposed sensor, as shown in Fig. 1. AG1 and AG2 can be separately expressed as

$$T_{\mathrm{AG1}} = \begin{bmatrix} \exp\left(-i\dfrac{2\pi L_1 n}{\lambda}\right) & 0 \\ 0 & \exp\left(-i\dfrac{2\pi L_1 n}{\lambda}\right) \end{bmatrix}$$

$$T_{\mathrm{AG2}} = \begin{bmatrix} \exp\left(-i\dfrac{2\pi L_2 n}{\lambda}\right) & 0 \\ 0 & \exp\left(-i\dfrac{2\pi L_2 n}{\lambda}\right) \end{bmatrix}$$

where L_1 and L_2 are the length of AG1 and AG2, respectively, and n is the RI of the chamber. The HiBi-PCF can be described as

$$T_{\mathrm{HiBi}} = \exp\left(-i\frac{2\pi n_e L_{\mathrm{HiBi}}}{\lambda}\right) \cdot \begin{bmatrix} 1 & 0 \\ 0 & \exp\left[-i\dfrac{2\pi L_{\mathrm{HiBi}}(n_o - n_e)}{\lambda}\right] \end{bmatrix}$$

where L_{HiBi} is the length of the HiBi-PCF, n_o and n_e are the refractive indices of ordinary and extraordinary light, respectively [16].

The transfer matrices of the clockwise and anticlockwise beams in the FLM can be written as

$$M_c = R_4 \cdot T_{\mathrm{AG2}} \cdot R_3 \cdot T_{\mathrm{HiBi}} \cdot R_2 \cdot T_{\mathrm{AG1}} \cdot R_1$$
$$M_a = R_1^{-1} \cdot T_{\mathrm{AG1}} \cdot R_2^{-1} T_{\mathrm{HiBi}} \cdot R_3^{-1} T_{\mathrm{AG2}} \cdot R_4^{-1}$$

in which

$$R_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{bmatrix}$$

where R_i denotes the rotation matrix of the polarization state induced by rotation angle θ_i at the end-face of SMF and HiBi-PCF when the light propagates clockwise.

We assume that port 1 of the coupler, shown in Fig. 1, has a input of E_1 which

$$E_1 = E_{\mathrm{in}} = \begin{bmatrix} E_{x\mathrm{in}} \\ E_{y\mathrm{in}} \end{bmatrix}$$

then after clockwise and anticlockwise propagation, the output at port 2 can be described as

$$E_2 = \sqrt{\frac{1}{2}} \cdot M_c \cdot E_3 + \sqrt{\frac{1}{2}} \cdot M_a \cdot H \cdot E_4 = K \cdot E_{\mathrm{in}}$$

in which

$$H = \begin{bmatrix} \exp[i(\pi/2)] & 0 \\ 0 & \exp[i(\pi/2)] \end{bmatrix}$$

$$K = \frac{1}{2} R_4 \cdot T_{\mathrm{AG2}} \cdot R_3 \cdot T_{\mathrm{HiBi}} \cdot R_2 \cdot T_{\mathrm{AG1}} \cdot R_1 + \frac{1}{2} R_1^{-1} \cdot T_{\mathrm{AG1}} \cdot R_2^{-1} T_{\mathrm{HiBi}} \cdot R_3^{-1} T_{\mathrm{AG2}} \cdot R_4^{-1} \cdot \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

where H is the phase delay caused by the coupler. Then we got the transmissivity of the proposed FLM,

$$T_{\mathrm{FLM}} = \frac{(K^* E_{\mathrm{in}}^*)^T K E_{\mathrm{in}}}{E_{\mathrm{in}}^* E_{\mathrm{in}}}.$$

In our experiments, we got the relationship between the transmission spectra and the RI of the chamber by simulation, shown in Fig. 8. In Fig. 8, $n = 1$ corresponds to the environment full of air, which is in the black line. While $n = 1.000878$ means only gaseous ethanol vapor existing in the chamber, which is in the red line. As Fig. 8 shown, the AGs only induce high frequency interferences while the sensitivity is not enhanced. So the influence of two AGs is also not the main reason contributing to the high sensitivity.

C. The Influence of Liquefaction

The discussions above demonstrate that the birefringence change ΔB is not induced by either EVC RI variation or AGs. However, the experiment results show an obvious wavelength

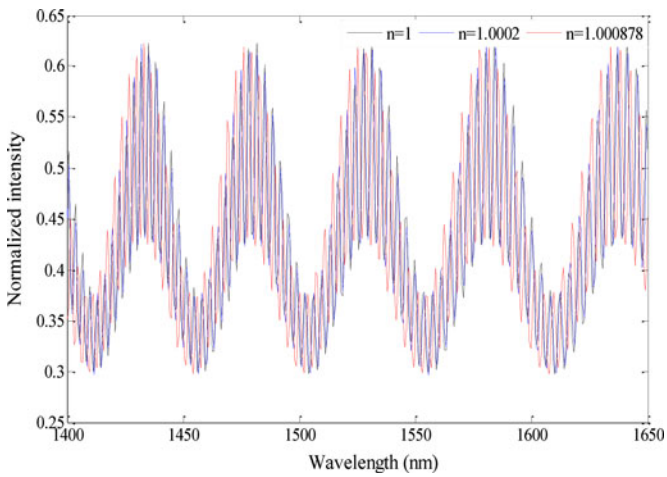

Fig. 8. Simulated spectra of the proposed chemical vapor sensor with different RI in the holes of HiBi-PCF.

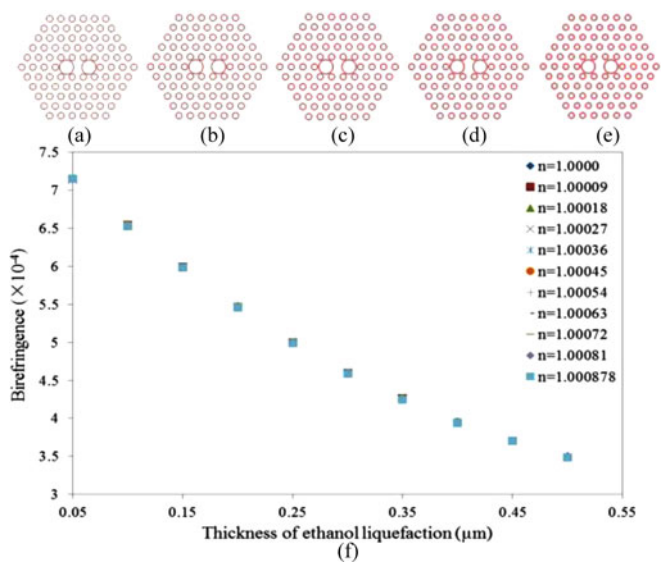

Fig. 9. (a)–(f) HiBi-PCF sections with different liquefaction thickness. (f) Relationships between the Birefringence and different liquefaction thicknesses.

shift. So a new model is proposed here, in which the liquefaction is considered. We assume that the gaseous ethanol vapor in the fiber holes will liquefy as the EVC increase in the chamber, and a liquefaction film will attach to the inside the fiber holes. As shown in Fig. 9(a)–(e), the birefringence simulations of the HiBi-PCF with different thickness of ethanol liquefaction film are achieved by using FEM. Top figures of Fig. 9 show the theoretical model of HiBi-PCF with different liquefaction thickness. For instance, Fig 9(a) shows the theoretical model of HiBi-PCF with 0.05 μm liquefaction thickness. The thicker the liquefaction thickness is, the darker the borderlines are. And Fig 9(e) shows the theoretical model of HiBi-PCF with 0.55 μm liquefaction thickness, which is in dark red borderlines. However, the change of the thickness is too little to observe clearly. The birefringence has an obvious change from 7.05×10^{-4} to 3.45×10^{-4} with a thickness of ethanol liquefaction film from 0.05 to 0.55 μm. The change trend becomes flat gradually with an increase of the thickness, shown in Fig. 9(f), which is similar to the trend of the experimental results, shown in Fig. 5. Moreover, the birefringence is irrelevant to the corresponding EVC, n, which is also shown in Fig. 9(f), meaning that the RI variation has no effect on the birefringence.

The simulation results indicate that the ethanol liquefaction contributes largely to the birefringence changes ΔB, which will cause an evident wavelength shift in the FLM transmission. According to the experimental results and Eq. (3), the extreme thin ethanol liquefaction of less than 0.05 μm can induce a birefringence change which achieves a relatively large blue shift referring to [8] and [10].

V. Conclusion

In conclusion, we have proposed a novel compact chemical vapor sensor based on HiBi-PCF FLM and analyzed the influence of three different mechanisms. Experimental results of the proposed HiBi-PCF FLM VOCs sensor show that the sensitivity for ethanol is as high as 15.5 pm/ppm. Three different mechanisms of the extremely high sensitivity are discussed, in which a liquefaction model is proposed.

In order to simplify the sensing instrument, a fiber capillary, of which the inner diameter, outer diameter is 127, 300 μm, respectively, can be used as a package to align the HiBi-PCF with SMFs instead of 3-D translation stages. Otherwise, the HiBi-PCF can be side-opened by using a femtosecond infrared laser [17] and fusion spliced with the SMFs so that the high frequency interference dips can be avoided. Although the proposed sensor is sensitive to VOC, it is difficult to recognize the target VOC because various types of vapor can liquefy and attach into the air holes of the PCF. In order to detect the target VOC, a material permeable, for instance, fabricating internally sol–gel coated the air holes of the PCF, can be applied to carry out the detection [19]. Further study will be needed.

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